JS EXES - basics

Section 1: Variable Swap Without Temp

Goal: Swap the values of two variables `a` and `b` without using a third variable.
Twist: The variables are strings.

example:

let a = “some string”;

let b = “other string”;

results should be:
a = “other string”
b = “some string”
without using a third variable to achieve this.

Section 2: 2D Grid Count

Goal: Count how many `"X"` values appear in a 5x5 2D array.
Twist: Ignore `"x"` (lowercase) and log coordinates of each `"X"`.

Section 3: Nested Object Read

Goal: Access and log the third child’s name from a nested object:

let family = { parents: {}, children: [{name: "Ali"}, {name: "Lea"}, {name: "Mona"}] };

Twist: Also print all sibling names in one comma-separated string.

Section 4: Manual Reverse

Goal: Reverse an array manually using a loop.
Twist: Do not use `.reverse()` or `.unshift()`.

Section 5: Type Collector

Goal: Given a mixed array, count how many of each type: `number`, `string`, `object`, `boolean`.
Twist: Also count how many values are `null`.

Section 6: Filter Without `.filter()`

Goal: From a number array, manually create a new array of numbers > 100.
Twist: Track how many comparisons were made.

Section 7: Conditional Object Filler

Goal: Create an object with keys `"even"` and `"odd"`, and push numbers 1–50 into the correct array.
Twist: Count how many are divisible by both 2 and 3.

Section 8: Grid Checker

Goal: Create a 3x3 grid using nested arrays and mark all diagonal values with `"#"`.
Twist: Mark both diagonals (top-left to bottom-right and vice versa).

Section 9: Frequency Counter

Goal: Given a string, count how many times each letter appears using an object.
Twist: Ignore spaces and punctuation, and make it case-insensitive.

Section 10: Flattening Matrix

Goal: Flatten a 2D number array into one 1D array.
Twist: Remove all negative numbers during the flattening.

Section 11: Object Key Transformer

Goal: Take an object and produce a new object where all keys are uppercase.
Twist: Also sort keys alphabetically before printing.

Section 12: Triangular Loop Output

Goal: Use nested loops to print a triangle made of `"*"` like this:

```
*
* *
* * *
```

Twist: Use variables to control size and symbol.

Section 13: Match Two Arrays

Goal: Print how many elements exist in both arrays A and B.
Twist: Arrays can have repeated values, but count matches once.

Section 14: Key-Value Reverser

Goal: Create a new object where the values become keys and the keys become values.
Twist: Only reverse if the value is a string or number.

Section 15: Identity Grid

Goal: Create a 5x5 array where cells with row === column contain `1`, others contain `0`.
Twist: Mark row 2 fully with `9`.

Section 16: Object Conditions

Goal: Given a list of user objects, print all users who are `"active"` and over 18.
Twist: Also count how many inactive users are under 18.

Section 17: Manual Min/Max

Goal: Find min and max values in an array using a loop.
Twist: Do not use `Math.min` or `Math.max`.

Section 18: Rotating Array

Goal: Rotate an array left by 1 step:

[1, 2, 3] → [2, 3, 1]

Twist: Repeat this 3 times in a loop.

Section 19: Manual JSON Print

Goal: Given an object, manually create a string that looks like JSON:

{ "name": "Ali", "age": 25 }
(add “ “ to each key)

Twist: No use of `JSON.stringify`.

Section 20: Complex Board Validator

Goal: Validate that a 4x4 game board has at most two `"O"`s in each row and column.
Twist: Print a message if the board violates the rule.